Schwarzschild died only a few months after finding his solution, from a skin disease he contracted at the Russian front. He was 42. His tragically brief encounter with Einstein's theory of gravity uncovered one of the most striking and mysterious facets of the natural world.

The second example in which general relativity flexes its muscle concerns the origin and evolution of the whole universe. As we have seen, Einstein showed that space and time respond to the presence of mass and energy. This distortion of spacetime affects the motion of other cosmic bodies moving in the vicinity of the resulting warps. In turn, the precise way in which these bodies move, by virtue of their own mass and energy, has a further effect on the warping of spacetime, which further affects the motion of the bodies, and on and on the interconnected cosmic dance goes. Through the equations of general relativity, equations rooted in geometrical insights into curved space spearheaded by the great nineteenth-century mathematician Georg Bernhard Riemann (more about Riemann later), Einstein was able to describe the mutual evolution of space, time, and matter quantitatively. To his great surprise, when the equations are applied beyond an isolated context within the universe, such as a planet or a comet orbiting a star, to the universe as a whole, a remarkable conclusion is reached: the overall size of the spatial universe must be changing in time. That is, either the fabric of the universe is stretching or it is shrinking, but it is not simply staying put. The equations of general relativity show this explicitly.

This conclusion was too much even for Einstein. He had overturned the collective intuition regarding the nature of space and time built up through everyday experiences over thousands of years, but the notion of an always existing, never changing universe was too ingrained for even this radical thinker to abandon. For this reason, Einstein revisited his equations and modified them by introducing something known as a cosmological constant, an additional term that allowed him to avoid this prediction and once again bask in the comfort of a static universe. However, 12 years later, through detailed measurements of distant galaxies, the American astronomer Edwin Hubble experimentally established that the universe is expanding. In a now-famous story in the annals of science, Einstein then returned to the original form of his equations, citing his temporary modification of